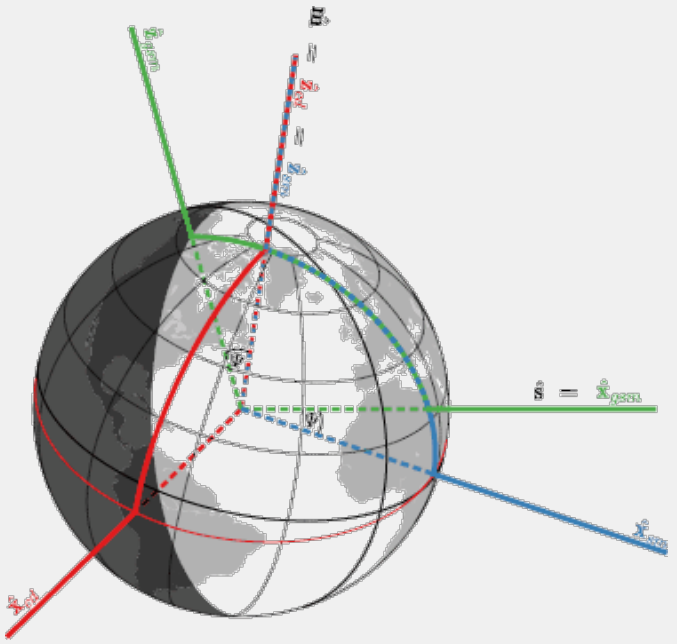
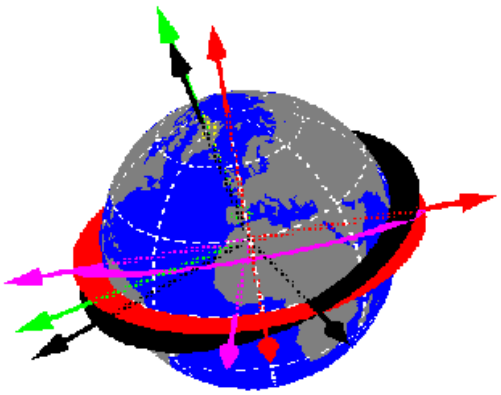
University of Colorado **Boulder**

# Magnetosphere: Beyond the Surface

# More on Coordinate Systems

- Geographic:
- Geodetic:
- GEI = Geocentric Equatorial Inertial:
- Geomagnetic:
- Geocentric Solar Ecliptic (GSE):
- Geocentric Solar Equatorial (GSEQ):
- Geocentric Solar Magnetospheric (GSM):
- Solar Magnetic (SM):
- Spacecraft:
- Transformation matrices map from one coordinate system to another





Axes

| | X | Y | Z |
|---|---|---|---|
| GEO | | | |
| GEI | | | |
| MAG | | | |
| SM | | | |
| GSM | | | |

Coordinates

Geodetic:

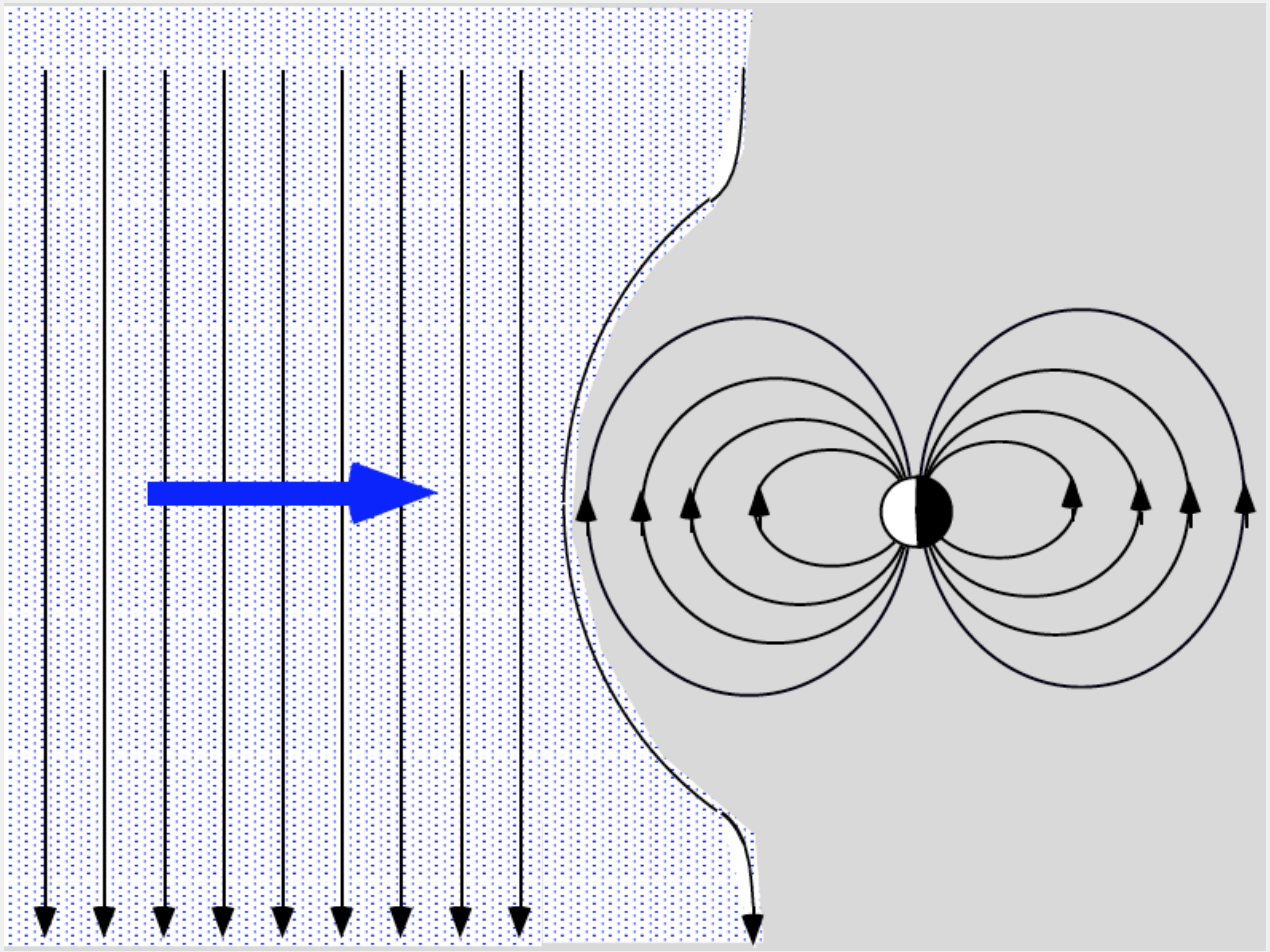
0.1504 50.7978 4.3592

Radius: 6365.5119

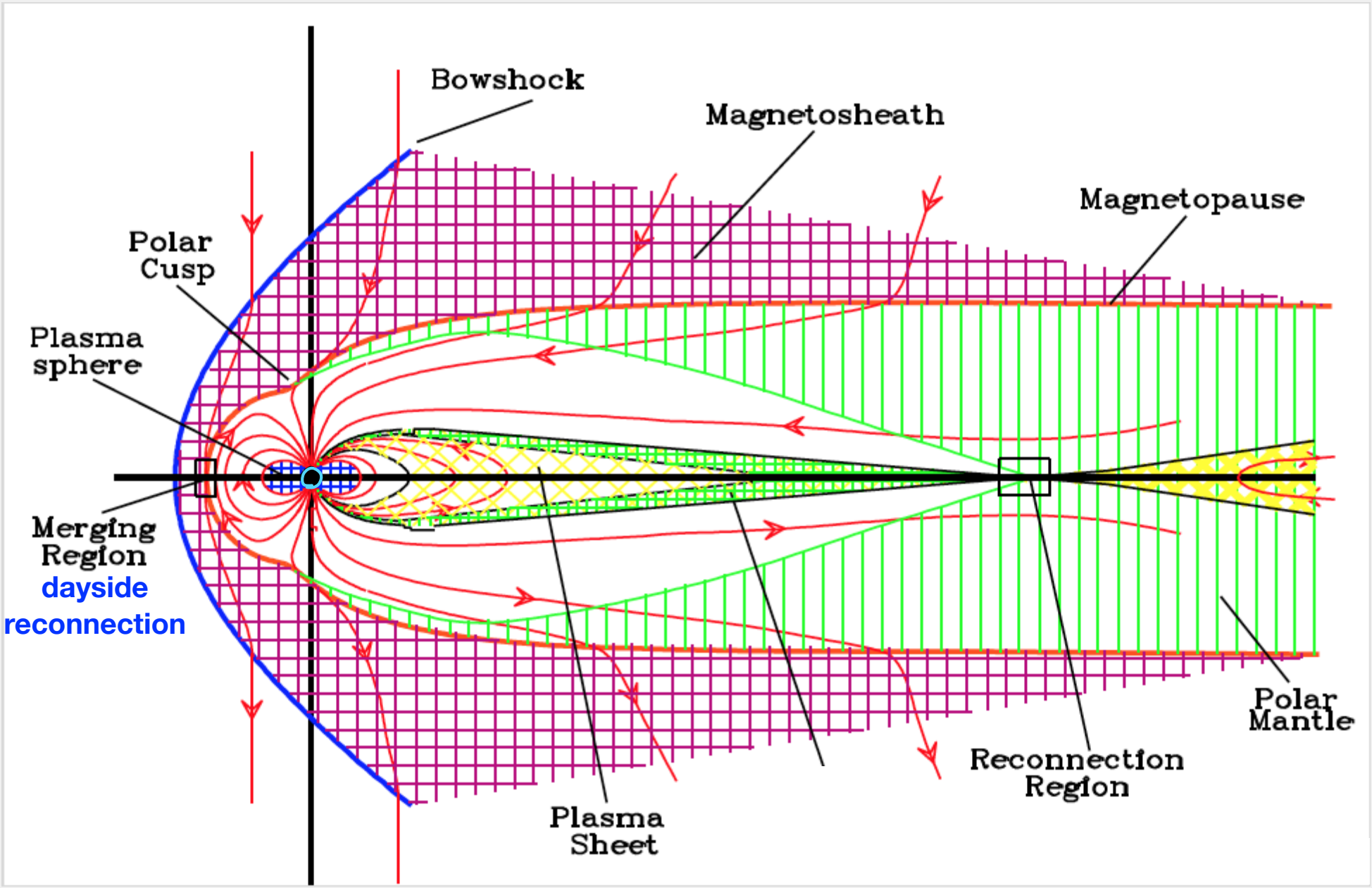
| | latitude | longitude |
|---|---|---|
| GEO | 50.6092 | 4.3592 |
| GEI | 50.6092 | 104.8044 |
| MAG | 51.9947 | 89.1577 |
| SM | 51.9947 | −167.9438 |
| GSM | 26.5415 | −171.7345 |
| $GEI_{J2000}$ | 50.6120 | 104.8169 |

# Magnetic Field in Space

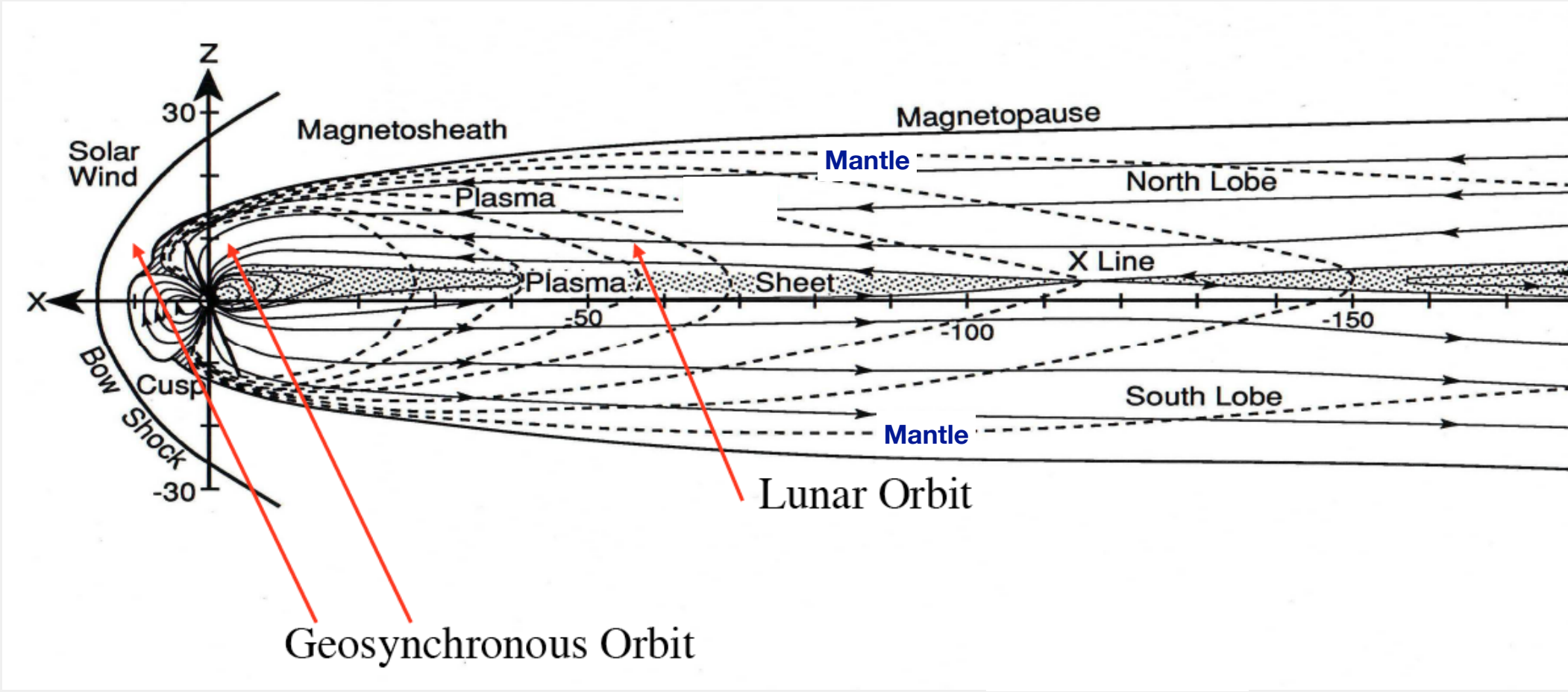
- As we get further out into space, the dipole (multipole) field model breaks down
- Field is heavily distorted by the solar wind
  - Solar wind carries particles (dynamic pressure) and drags the interplanetary magnetic field (IMF)



# Magnetosphere

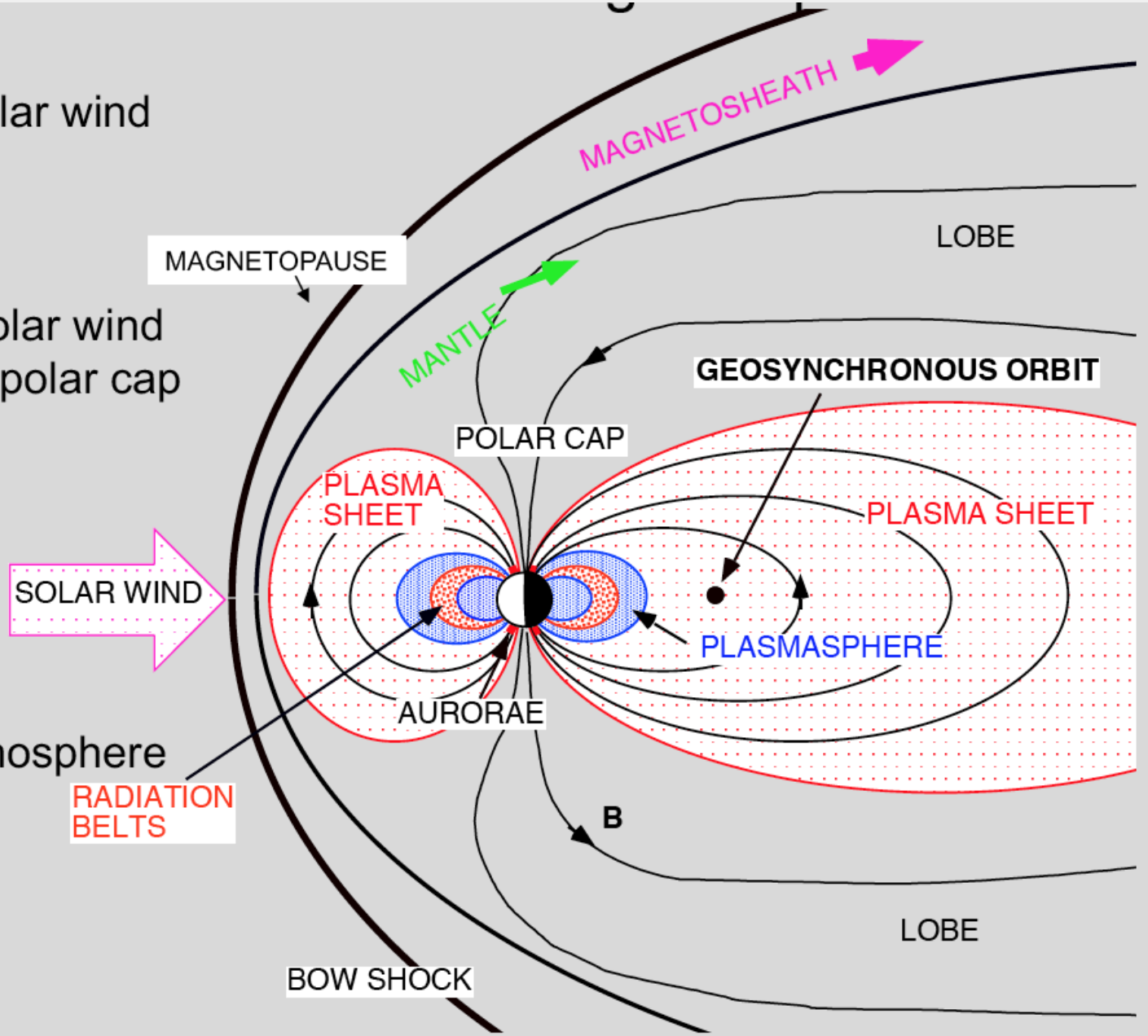


# Magnetosphere to Scale



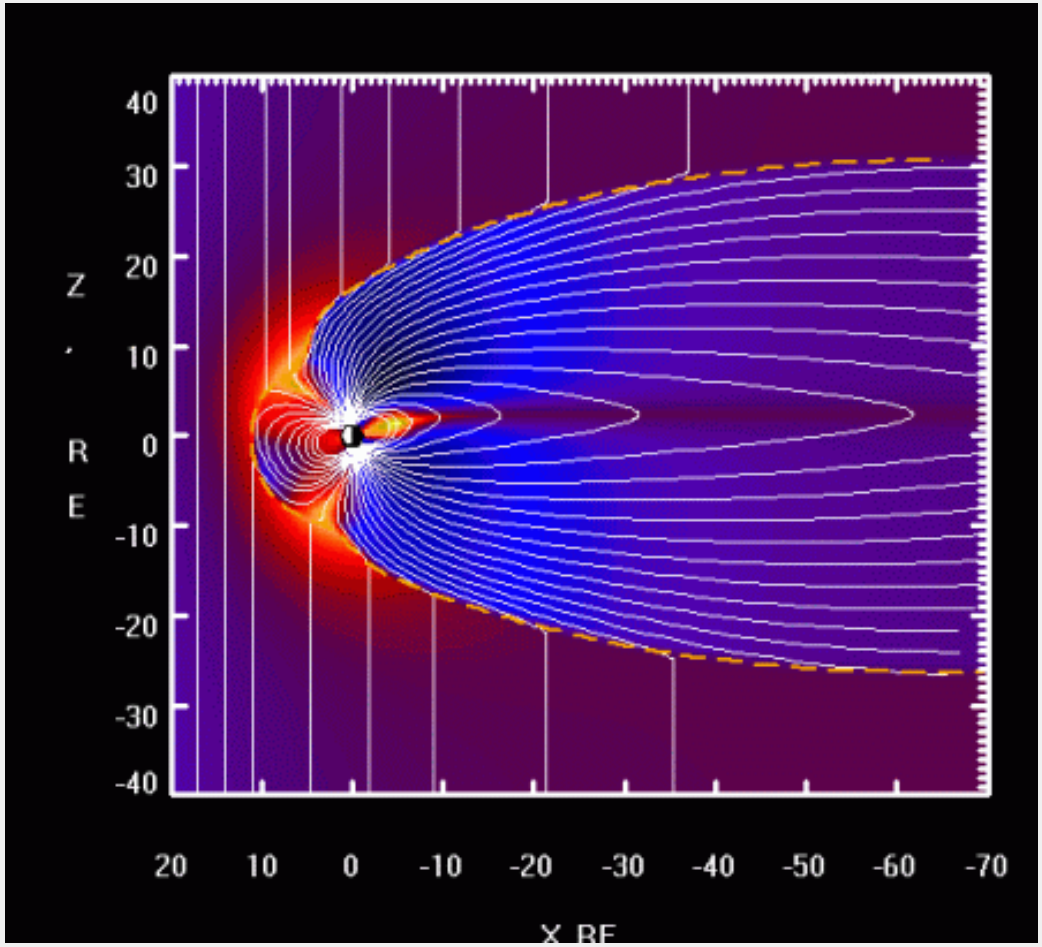
# Basic Structure of (Inner) Magnetosphere

- Magnetosheath
  - Compressed solar wind
- Mantle
  - Outer cloak
- Lobe
  - Connected to solar wind downtail and to polar cap at Earth
  - Cool
- Plasmasheet
  - Flexible dipole
  - Warm
- Plasmasphere
  - Extension of Ionosphere
  - Cold
- Radiation Belts
  - Mostly Dipole
  - Hot



# How you could possibly model this mess?

- But Nikolai Tsyganenko (University of St. Petersburg) is smart!
- https://ccmc.gsfc.nasa.gov/models/modelinfo.php?model=Tsyganenko%20Magnetic%20Field
- semi-empirical fit to a whole ton of spacecraft magnetometer data: IMP, HEOS, ISEE, POLAR, Geotail, GOES



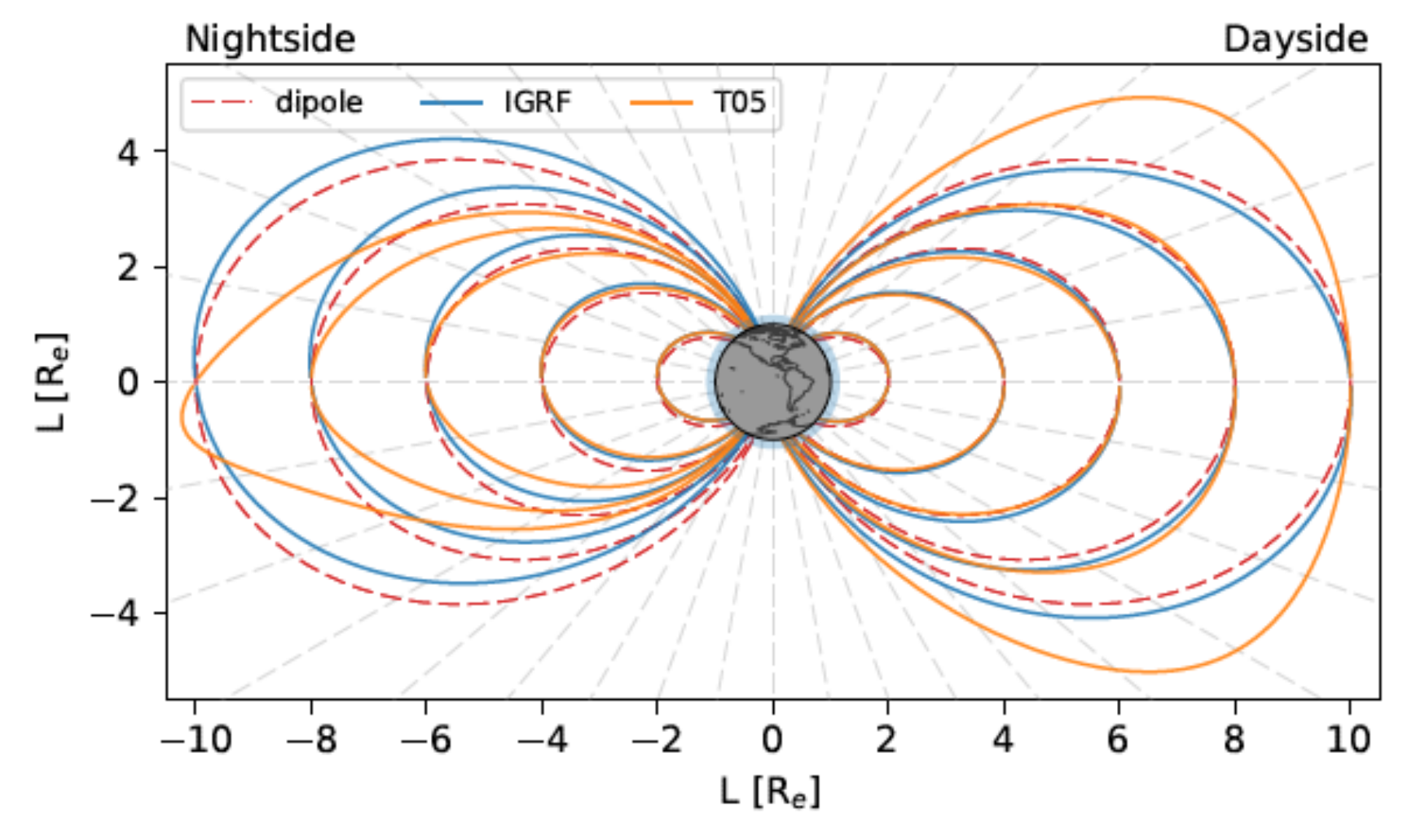





- Model updated every so often: T89, T96, T01, T02, TS07 (*S is for Sitnov*)
- TS07 continually updated with new data

# Comparison of Models in Inner Magnetosphere

- Figure courtesy Austin Sousa (CU LAIR)
- Dipole / IGRF models do pretty well inside L ~ 6, not so well outside





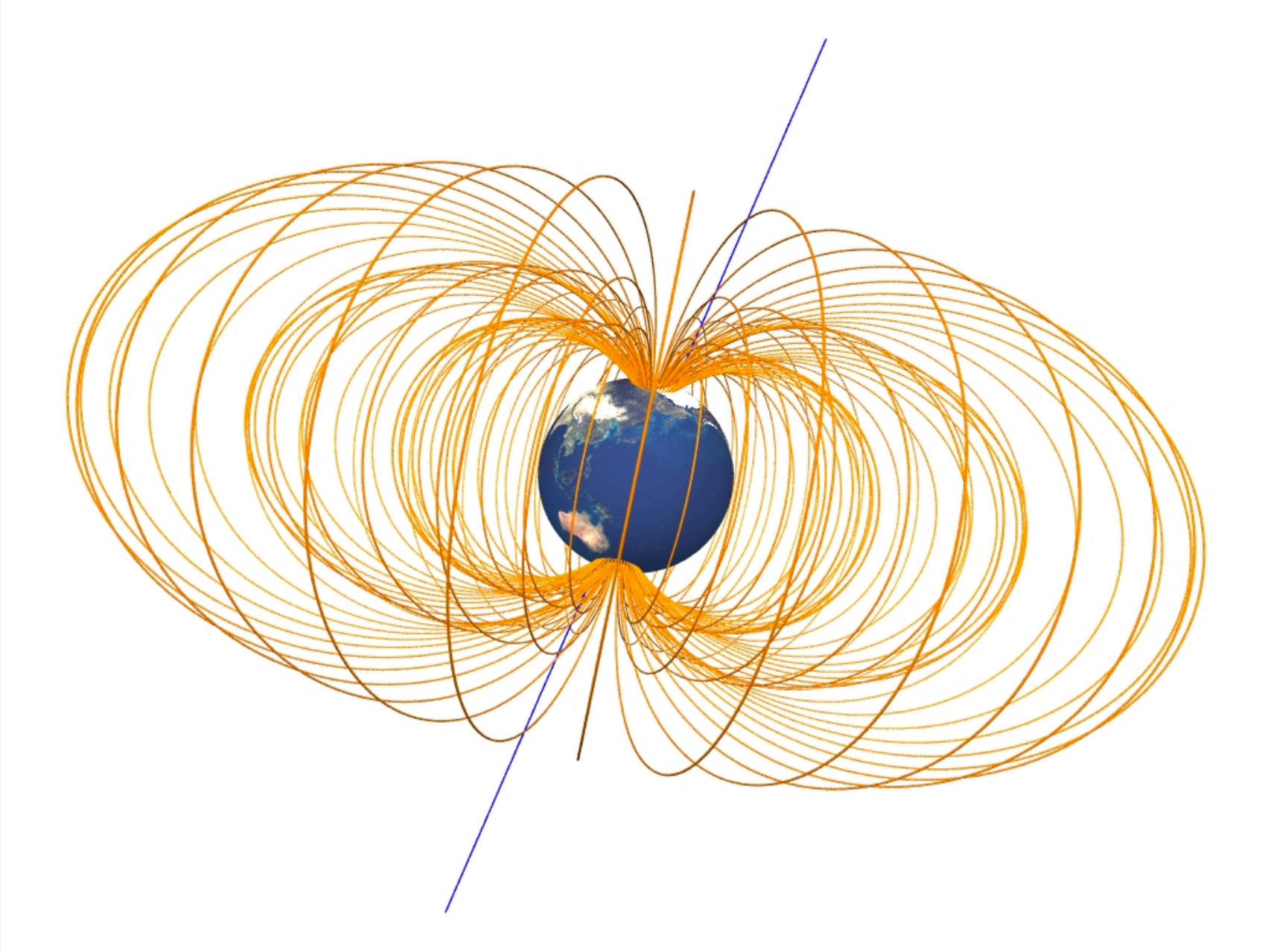
University of Colorado **Boulder**

# Charged Particle Motion in Magnetic Fields

# Lorentz Force: Gyromotion

# Gyromotion

# What does this look like in the magnetosphere?



# How fast are particles spinning?

- Gyrofrequency only depends on B and mass
- Gyroradius also depends on velocity (i.e. energy)

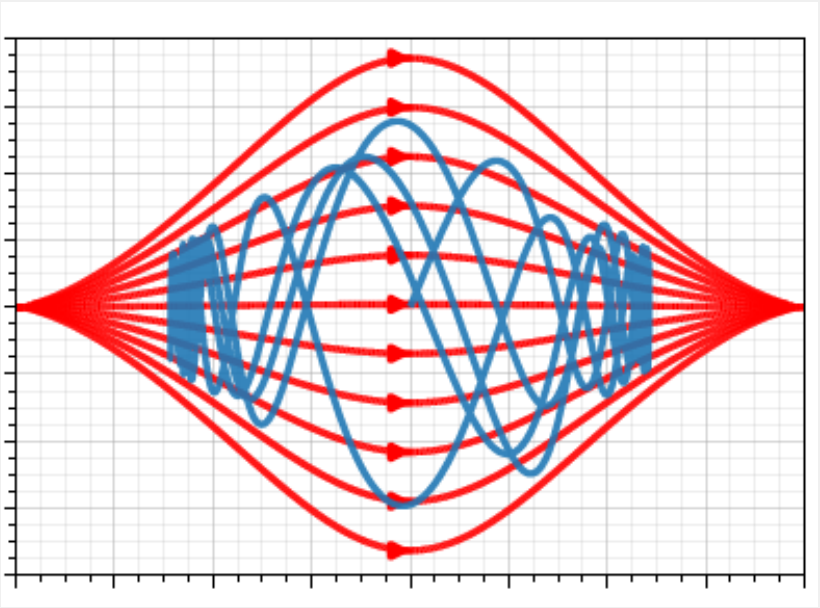
| $\lvert B\rvert =$ | 30,000 nT | 3,000 nT | 300 nT |
|---|---|---|---|
| $f_{ce}$ (electron) | 840 kHz | 84 kHz | 8.4 kHz |
| $r_g$ (1 eV electron) | 11 cm | 1.13 m | 11.3 m |
| $r_g$ (1 keV electron) | 3.56 m | 35.6 m | 355 m |
| $r_g$ (1 MeV electron) | 53.6 m | 536 m | 5.36 km |
| $f_{cp}$ (proton) | 457 Hz | 45.7 Hz | 4.57 Hz |
| $r_g$ (1 eV proton) | 2.6 mm | 2.6 cm | 26 cm m |
| $r_g$ (1 MeV proton) | 2.63 m | 26.3 m | 263 m |
| $r_g$ (100 MeV proton) | 24.4 m | 244 m | 2.44 km |
| $f_{cO}$ ($O^+$) | 28.7 Hz | 2.87 Hz | 0.287 Hz |
| $r_g$ (0.1 eV $O^+$) | 0.2 mm | 2.1 mm | 2.1 cm |
| $r_g$ (1 eV $O^+$) | 0.7 mm | 6.6 mm | 6.6 cm |

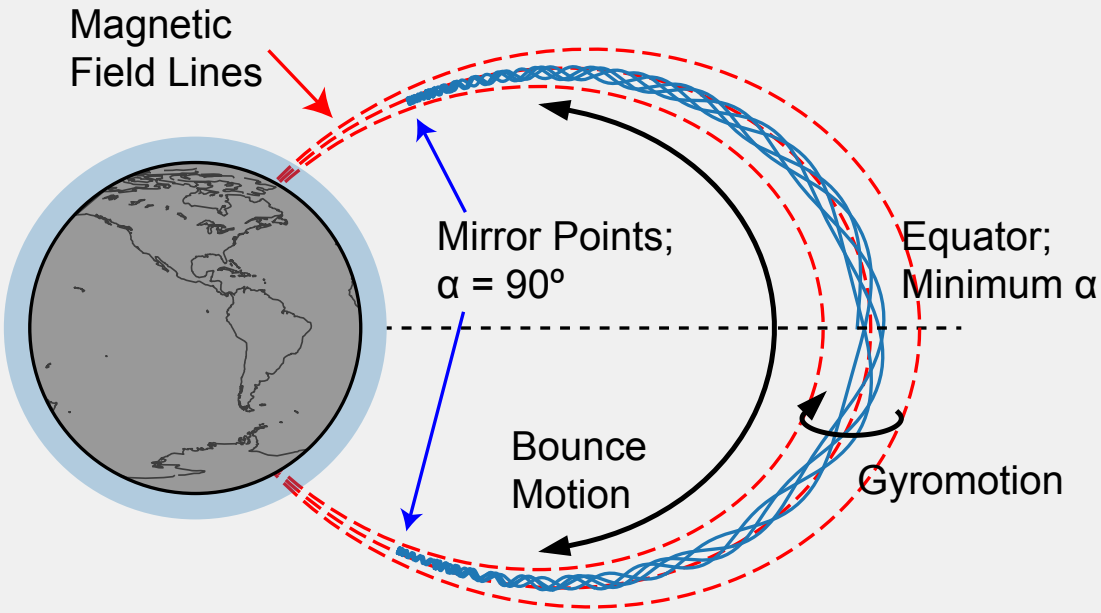
# Pitch Angle

- $v_\perp$ is affected by B-field; what about $v_\parallel$?
- **Pitch angle** defines spiral motion around field line

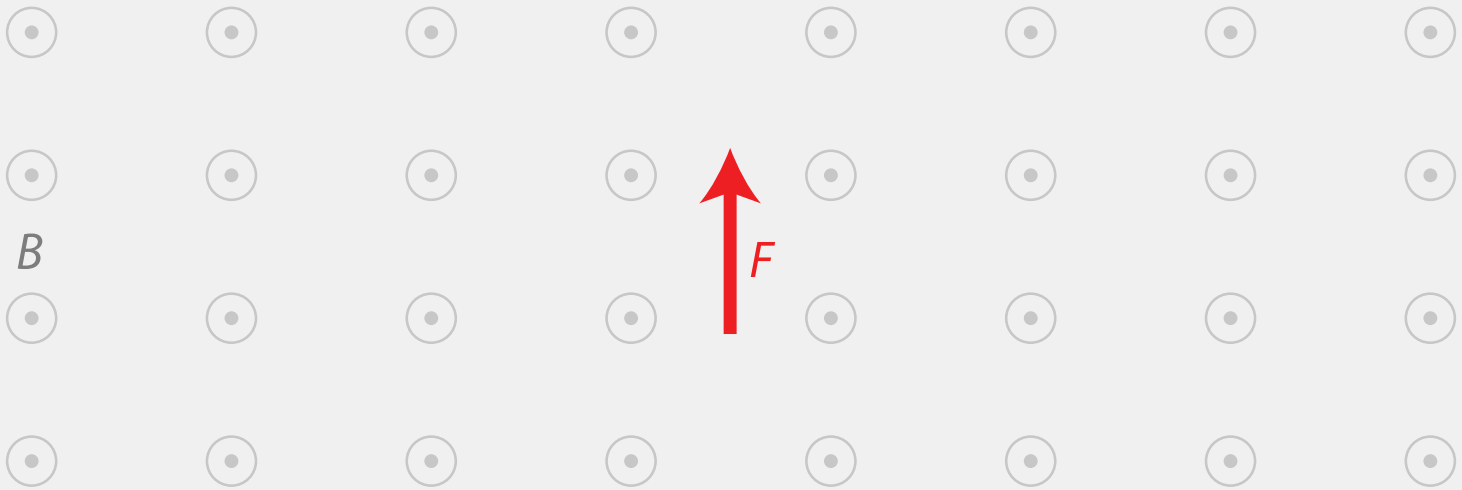
- What happens when particle goes into stronger field?
  - gyro frequency increases;
  - gyro radius decreases;
  - but velocity doesn't change;
  - instead, energy gets transferred from $v_\parallel$ to $v_\perp$

# Magnetic Mirror (or Bottle) => bounce motion



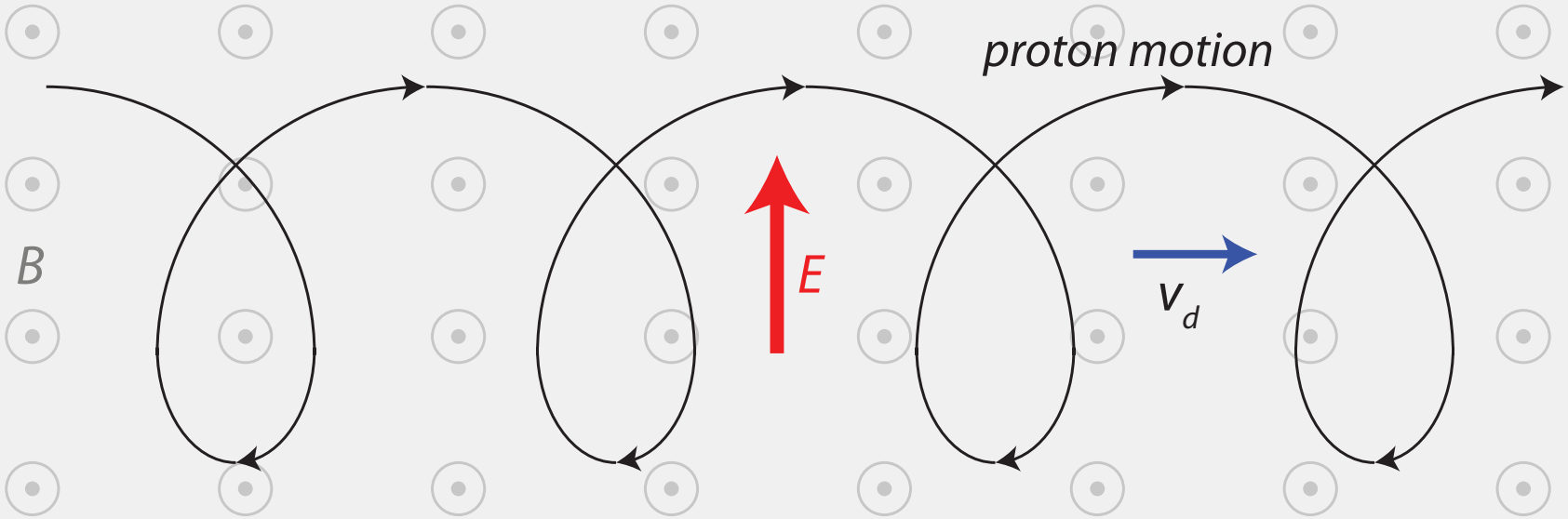


# Third Motion: Particle Drifts



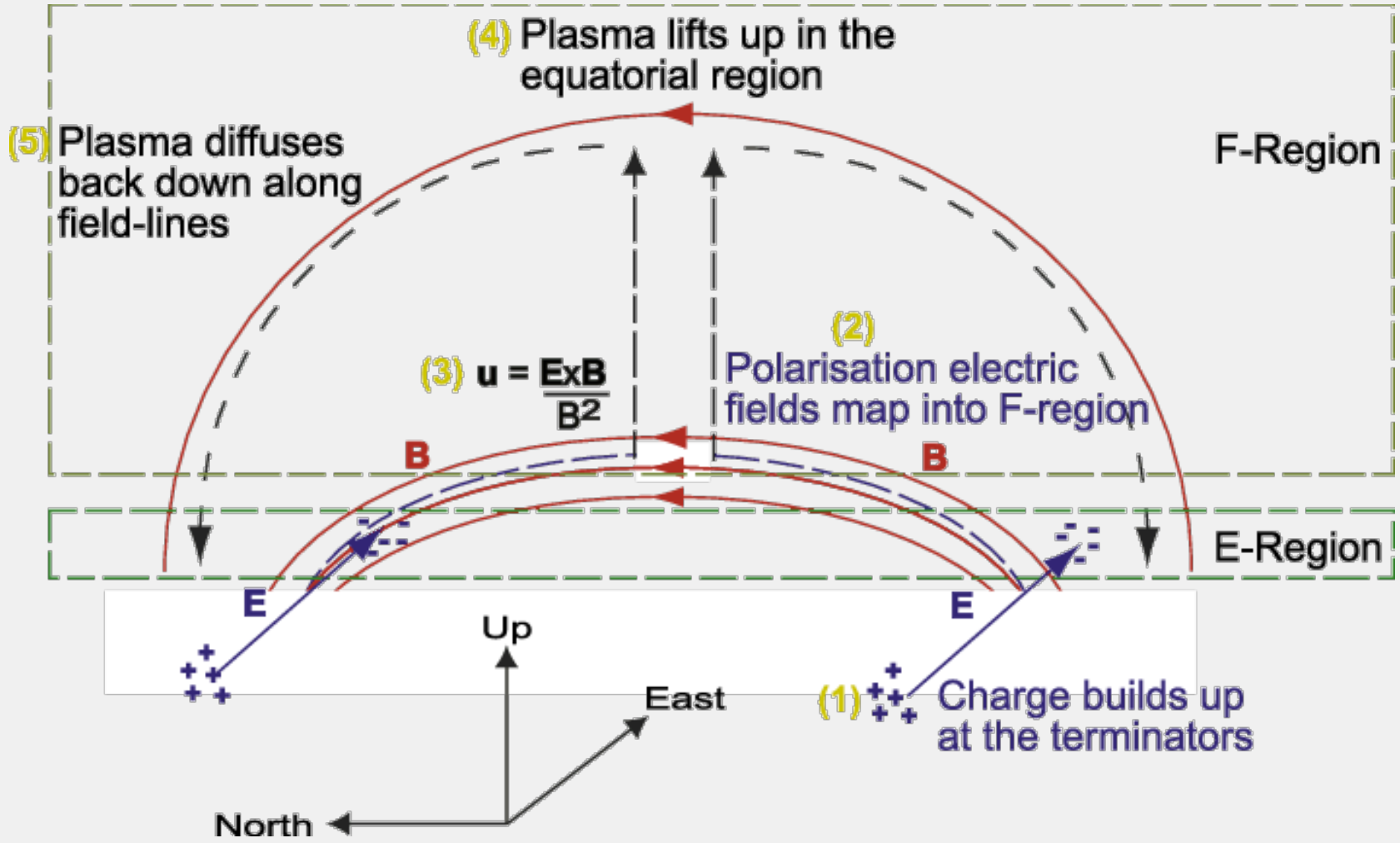
# Example: E x B drift

- What happens to particle in perpendicular E and B fields?

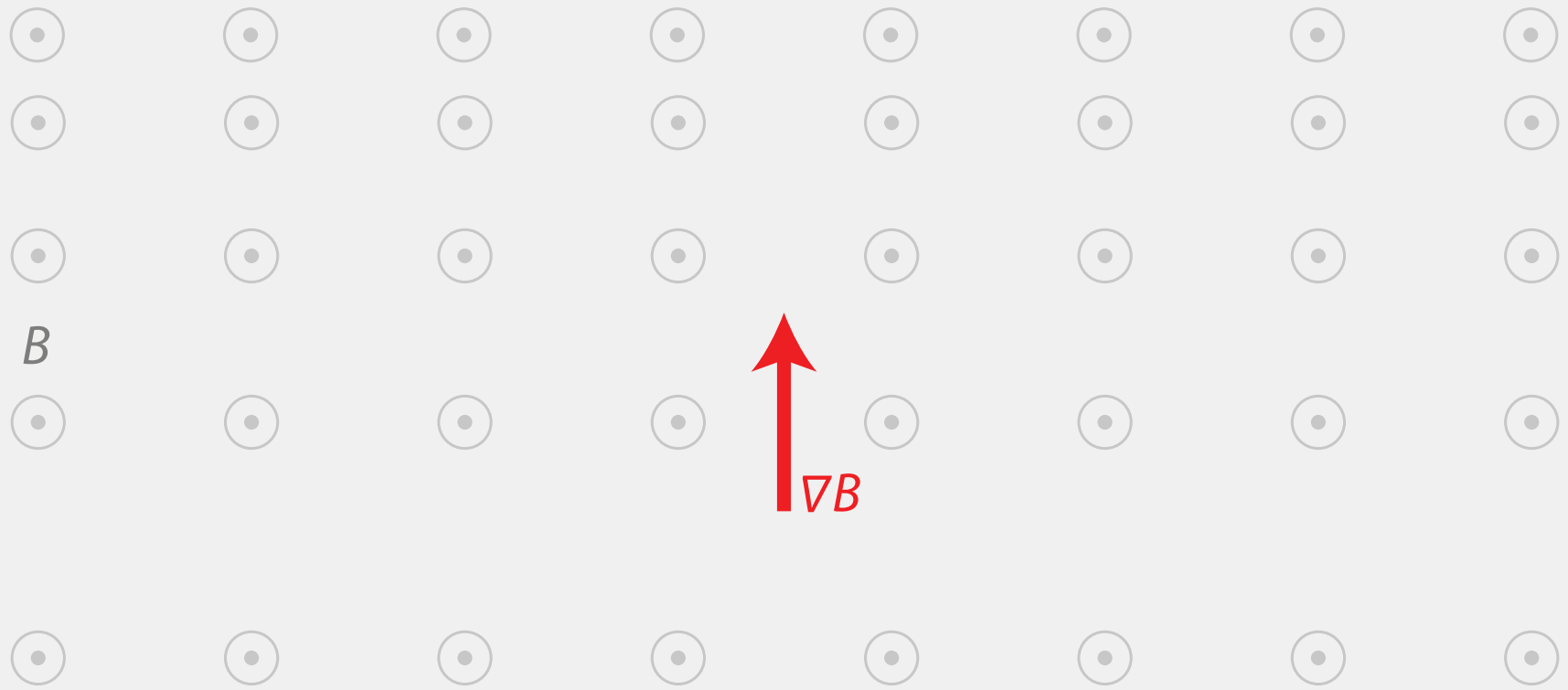


# Revisit: The Fountain Effect (ionosphere)

- Recall: E pointed **into the page**; B to the **left**; drift velocity **up**



# Gradient Drift



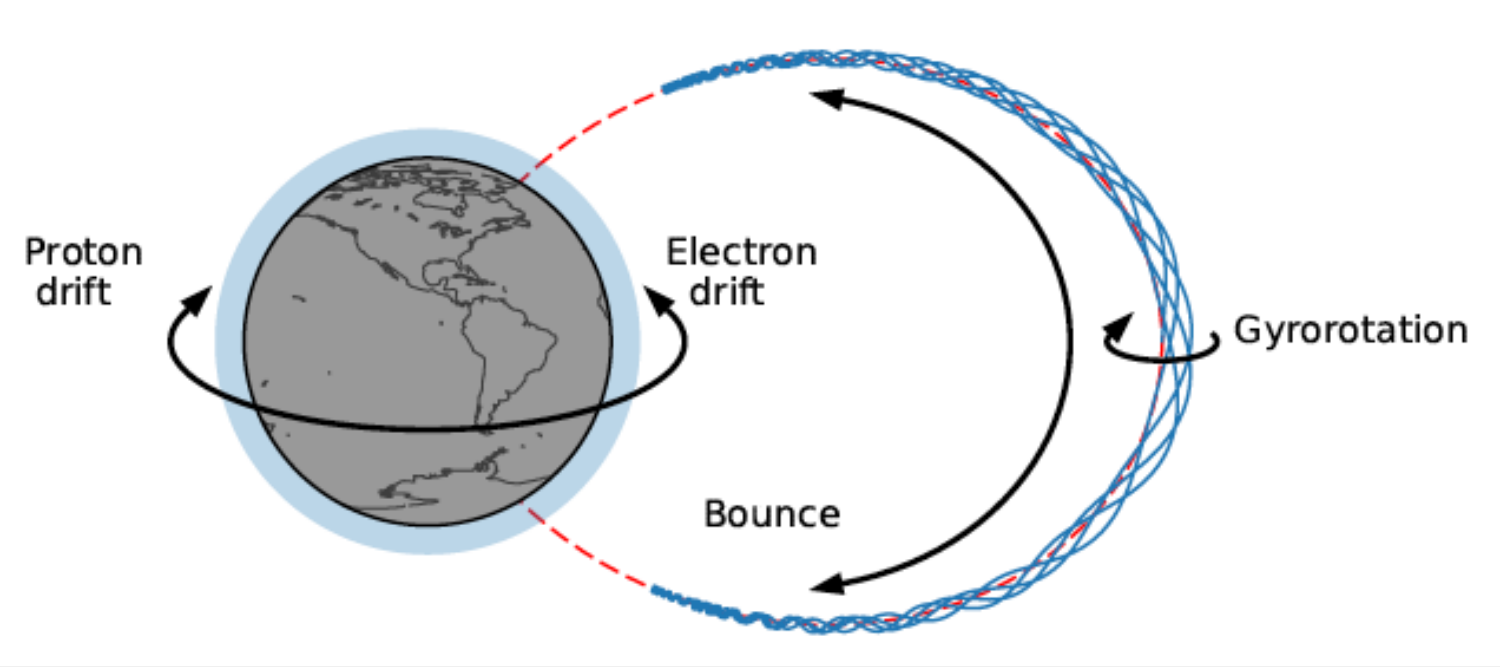
# Gradient-B Drift and Curvature Drift, in Math

(not considering the relativistic effects)

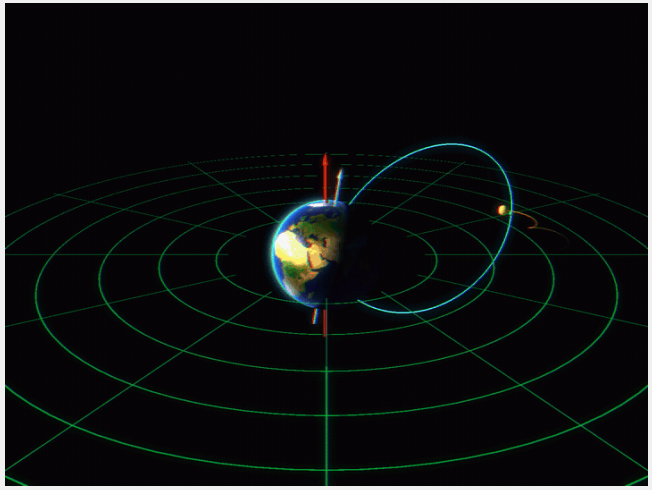
| $L =$ | 1.5 | 3 | 6 |
|---|---|---|---|
| 1 keV | 488 hours | 244 h | 122 h |
| 100 keV | 4.88 h | 2.44 h | 1.22 h |
| 1 MeV | 29.3 min | 14.6 min | 7.3 min |
| 100 MeV | 17.6 sec | 8.8 sec | 4.4 sec |

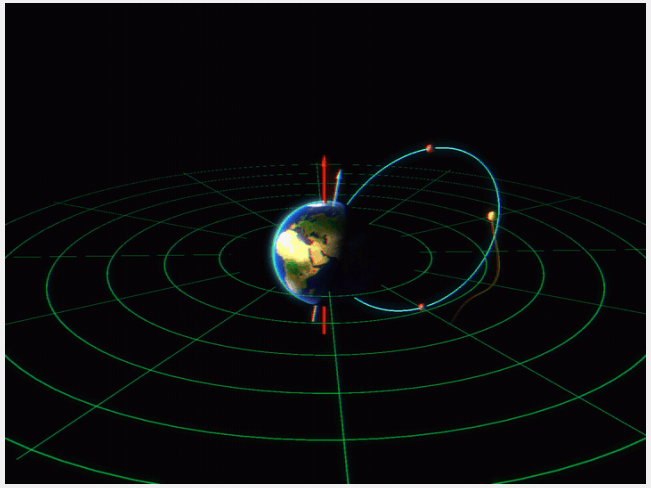
# Summary on Particle Motions in Magnetic Field

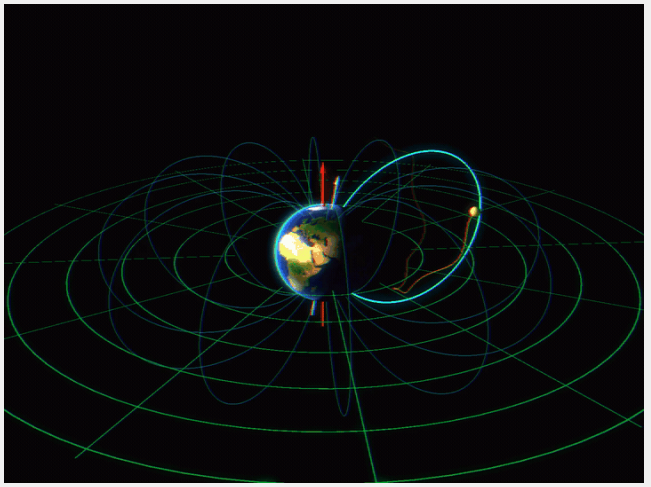
- Three types of *adiabatic* motion:
  - **gyromotion** around field lines
  - **bounce motion** in magnetic mirror
  - **drift motion** in curved fields

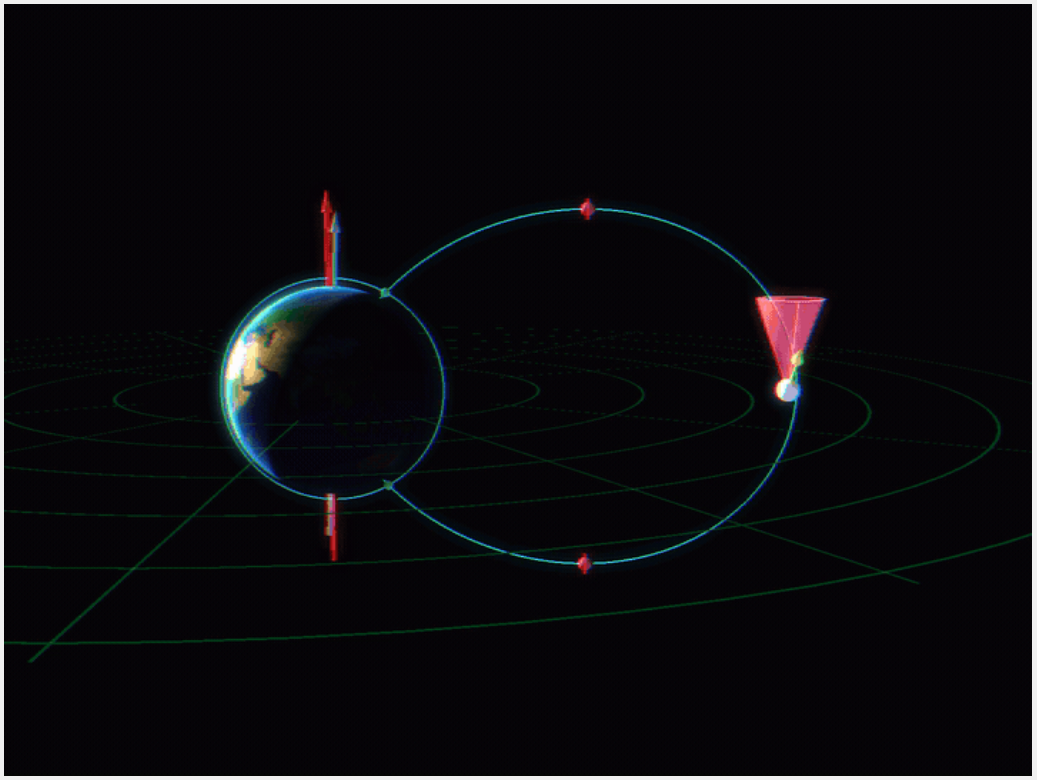


# Charged Particle Motion in Magnetosphere



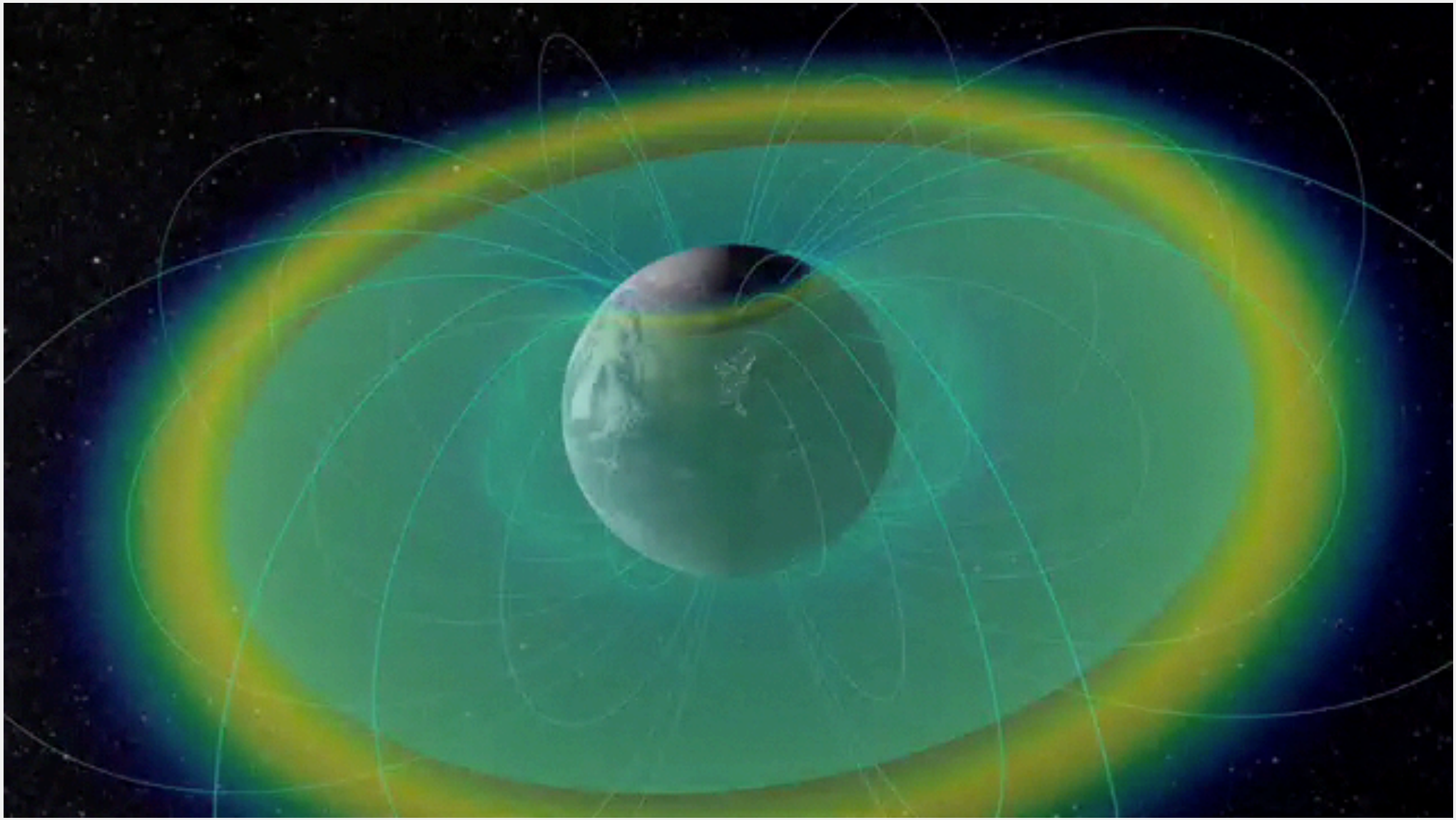




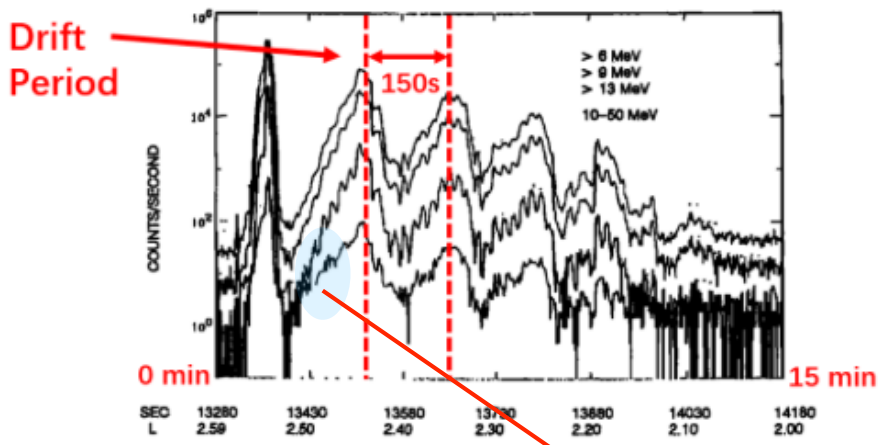


Courtesy of Breno Massacre

# Charged Particle Motion in Magnetosphere

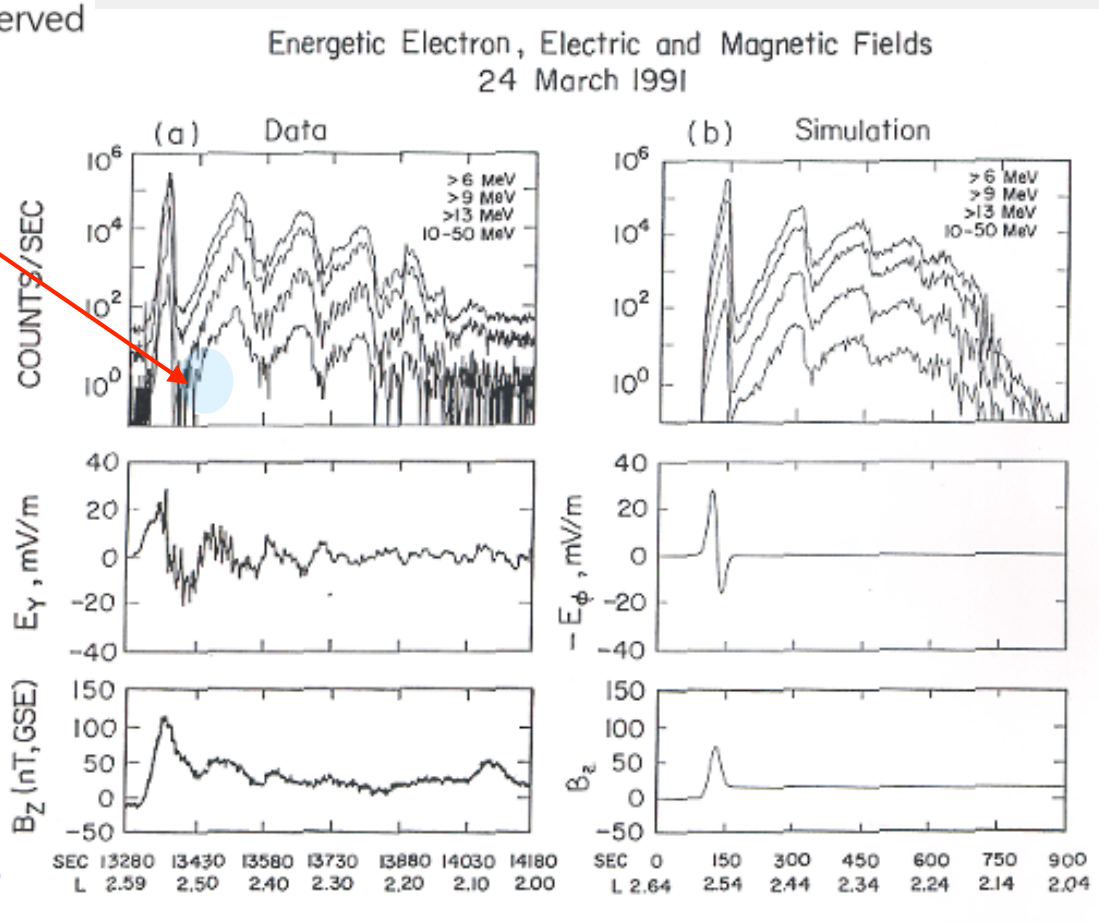


# A unprecedented (and fortuitous) event stunned the Community




One of the Most Dramatic Radiation-Belt Injections Ever Observed

**CRRES** Blake et al., 1992




Li et al., GRL, 1993



University of Colorado **Boulder**

# Particles and Currents in the Magnetosphere

# Basic Structure of (Inner) Magnetosphere

